DS-UA 202, Responsible Data Science, Spring 2026
Technical Audit of an ADS
Final Report

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5/11/2026

1 Background

1.1 Purpose and Stated Goals

The ADS to be analyzed [3] originates from a Kaggle competition called *Predicting Loan Payback* [1].

We selected this ADS as we noticed that many models in this specific Kaggle competition primarily optimize accuracy. However, in this course, the fairness of algorithmic decisions has also been an important concern historically, although it is sometimes overlooked in practice. In addition to traditional performance metrics such as accuracy, there are established fairness metrics and evaluation methods designed to assess how model decisions affect different groups. In this report, we apply these fairness metrics and evaluation methods to reassess the overall performance of this ADS.

1.2 Trade-offs Between Goals

Our chosen ADS is intended to improve the efficiency and consistency of credit-risk assessment. Rather than relying on manual review, the model try to use data driven predictions to support decisions on whether a loan should be approved or denied. In the notebook, the objective is to achieve strong predictive performance evaluated by metrics like accuracy and AUC. More broadly, the system is designed to forecast loan repayment outcomes and assist lending decisions.

When an ADS pursues multiple goals simultaneously, tension between those goals can emerge. We identify two key trade-offs in this system.

First, there is a trade-off between predictive accuracy and fairness. A model optimized only for strong performance may rely on features that create unequal outcomes across different demographic groups.

Second, there is a trade-off between model performance and interpretability. The ADS uses a 10-fold ensemble of LightGBM and CatBoost. This kind of complex ensemble may achieve better predictive performance, but it sacrifices interpretability in the process.

2 Input and Output

2.1 Data Collection and Selection

We obtained the dataset [2], both training and test splits, from the Kaggle competition *Predicting Loan Payback* [1]. The dataset for this competition was generated from a deep learning model trained on the Loan Prediction dataset [4].

2.2 Input Feature Profiling

All features in this dataset have no missing values. Below we describe each feature's data type and distribution.

- **annual_income** (float64): The distribution is bimodal and positively skewed, centered around \$50,000. The two modes are approximately \$25,000 and \$50,000.
- **debt_to_income_ratio** (float64): The distribution is positively skewed, centered between 0.1 and 0.2.
- **credit_score** (int64): The distribution is approximately normal with a left tail from borrowers with scores below 500. The center is around 680.
- **loan_amount** (float64): The distribution is approximately normal with a thick right tail from borrowers with loans above \$32,000. The center is around \$15,000.
- **interest_rate** (float64): The distribution is approximately normal with balanced tails, centered around 12.5.
- **gender** (Object): There are slightly more female than male borrowers in the dataset, with a small number of non-binary individuals.
- **marital_status** (Object): Single and married borrowers are the majority, with slightly more single individuals. There are also small numbers of divorced and widowed individuals.
- **education_level** (Object): The most common education level is Bachelor's degree, followed by High School, Master's degree, Other, and PhD.
- **employment_status** (Object): Employed is the dominant category.
- **loan_purpose** (Object): Most borrowers use loans for debt consolidation.
- **grade_subgrade** (Object): This represents the risk category assigned to a loan. Most borrowers fall into letter grades C and D, indicating moderate to low risk.
- **loan_paid_back** (float64): This binary target variable appears only in the training set. A value of 1 indicates that the borrower fully repays the loan; a value of 0 indicates that the borrower does not.

2.2.1 Pairwise Correlations

	annual_income	debt_to_income	credit_score	loan_amount	interest_rate	loan_paid_back
annual_income	1.0000	0.0008	0.0006	-0.0029	-0.0018	0.0064
debt_to_income	0.0008	1.0000	-0.0619	-0.0081	0.0299	-0.3355
credit_score	0.0006	-0.0619	1.0000	-0.0064	-0.5379	0.2341
loan_amount	-0.0029	-0.0081	-0.0064	1.0000	0.0005	-0.0032
interest_rate	-0.0018	0.0299	-0.5379	0.0005	1.0000	-0.1293
loan_paid_back	0.0064	-0.3355	0.2341	-0.0032	-0.1293	1.0000

Table 1: Correlation Matrix of Loan Dataset Variables

The most notable correlations are between `debt_to_income_ratio` and `loan_paid_back` ($r = -0.3355$), and between `interest_rate` and `credit_score` ($r = -0.5379$). All other pairwise correlations are small.

2.3 Output

The output of this ADS is a binary label indicating whether a loan borrower will fully repay the loan. A value of 1 indicates full repayment; a value of 0 indicates non-repayment. This label is provided in the training dataset as the variable `loan_paid_back`, and it is the quantity the model predicts on the test set.

3 Implementation and Validation

3.1 Data Cleaning and Pre-processing

The dataset is synthetically generated and there is not null values in any row for any column. Let's just talk about the feature-engineering step.

The original model author chooses to do ordinal encoding on `'education_level'` and one-hot encoding on `'gender'`,`'marital_status'`,`'employment_status'`. Additionally, the author engineers several new features from existing columns, and later adds `risk_factor`, `credit_efficiency`, and `dti_grade` during the hyperparameter tuning phase. A notable concern across the encoding step is that `grade_subgrade` and `loan_purpose` are mapped separately on the training and test sets, meaning the same category can receive different numeric values in each split, which is a potential source of inconsistency at inference time.

The original values of columns that the author wants to encode:

```
education_level: Bachelor's, High School, Master's, Other, PhD.
gender: Female, Male, Other.
marital_status: Single, Married, Divorced, Widowed.
employment_status: Employed, Unemployed, Self-employed, Retired, Student.
```

3.1.1 Concerns about Encoding education_level

Currently, the author encodes `'education_level'` based on the following ordinal schema:

```
Other:0,
High School:1,
Bachelor's: 2,
Master's: 3,
PhD: 4
```

This way of ordinal encoding relies on the assumption that the relationship between education level and loan repayment is monotonic. This may not hold because of two potential reasons. First, higher-educated borrowers often carry significant student debt, which could reduce repayment likelihood for additional loans. Second, the relationship between education and repayment may be non-monotonic. For example, people who just graduated from a PhD program might behave more like high-schoolers than Master's holders in terms of debt burden. However, our arguments also relies on the assumed situation stated above, so we will calculate repayment rate and visualize load purpose for each education level.

The table below presents the repayment statistics (sorted by the loan paid back rate) for each education level group, including total borrower count, number of loans repaid, mean repayment rate, variance, and standard deviation.

(Encoding) Edu_level	Total	Paid Back	Mean	Var	Std
(2) Bachelor's	279606	220579	0.7889	0.1665	0.4081
(3) Master's	93097	74696	0.8023	0.1586	0.3982
(0) Other	26677	21416	0.8028	0.1583	0.3979
(1) High School	183592	148654	0.8097	0.1541	0.3925
(4) PhD	11022	9149	0.8301	0.1411	0.3756

Table 2: Repayment Statistics by Education Level (sorted by mean repayment rate)

The repayment rate across education levels **is not monotonically increasing** with the assigned ordinal values, which empirically undermines the justification for ordinal encoding. Notably, Bachelor's degree holders exhibit the lowest repayment rate (0.789), consistent with our argument that student debt burden may reduce repayment likelihood for additional loans. Surprisingly, the "Other" category achieves a relatively high repayment rate (0.803), comparable to Master's degree holders. We suspect this group may be a mixture of individuals with education levels below High School and above PhD (e.g., postdoctoral researchers), which could produce a heterogeneous but not necessarily low-repaying subpopulation. However, the standard deviation for “Other” (0.398) is not substantially higher than other groups, which does not strongly support our mixing hypothesis. But the “Other” group also have considerably smaller sample size (26677), suggesting that its statistics are less reliable estimates of the true population parameters. This small sample of “Other” also makes sense because most education levels are already covered by other variables, leaving a smaller subgroup in the population for “Other” to cover.

The table below presents the distribution of loan purposes across different education levels.

(Encoding) Edu_level	Business	Car	Debt Consol.	Education	Home	Medical	Other	Vacation
Other	0.0607	0.1072	0.5477	0.0562	0.0723	0.0375	0.1056	0.0128
High School	0.0591	0.0966	0.5496	0.0592	0.0733	0.0382	0.1090	0.0151
Bachelor’s	0.0598	0.0974	0.5428	0.0644	0.0748	0.0390	0.1073	0.0144
Master’s	0.0573	0.1018	0.5484	0.0604	0.0764	0.0373	0.1057	0.0127
PhD	0.0702	0.0738	0.5764	0.0587	0.0641	0.0361	0.1101	0.0105

Table 3: Loan Purpose Distribution by Education Level (row-wise proportions)

All loan purposes apart from debt consolidation show small but notable differences across groups. Bachelor’s degree holders have the highest proportion of education loans (6.4%), which is consistent with the hypothesis that this group carries relatively more education-related debt burden. Medical and vacation loans are consistently low and similar across all groups. However, a key limitation of this analysis is debt consolidation. It combines different kinds of debts, which obscures the original source of debt. A borrower consolidating student loans is indistinguishable from one consolidating credit card or medical debt, meaning that education-driven debt burden cannot be confirmed or ruled out from this feature alone. Overall, while minor differences in borrowing patterns exist across education levels, they are not substantial enough to fully explain the non-monotonic repayment pattern observed earlier, suggesting that other financial factors such as debt_to_income_ratio and credit_score are likely more determinative.

3.1.2 Encoding other features in this dataset

The one-hot encoding `gender, marital_status, and employment_status` is appropriate, since those variables are nominal and categorical with no natural ordering, and the one-hot encoding makes no assumptions about the relationships between categories.

In addition to encoding existing features, the author engineers several new features from existing columns. `loan_to_income` is computed as the ratio of `loan_amount` to `annual_income`, capturing the relative borrowing burden. `risk_score` is the product of `debt_to_income_ratio` and `interest_rate`, combining two financial stress indicators into a single value. `log_income` applies a log transformation to `annual_income` to compress its skewed distribution. `grade_main` is derived by integer-dividing `grade_subgrade` by 5, effectively extracting the letter grade while discarding the sub-grade distinction.

After feature engineering, the training set contains 21 features: `annual_income`, `debt_to_income_ratio`, `credit_score`, `loan_amount`, `interest_rate`, `education_level`, `loan_purpose`, `grade_subgrade`, `gender_Male`, `gender_Other`, `marital_status_Married`, `marital_status_Single`, `marital_status_Widowed`, `employment_status_Retired`, `employment_status_Self-employed`, `employment_status_Student`, `employment_status_Unemployed`, `loan_to_income`, `risk_score`, `log_income`, and `grade_main`.

3.2 Implementation Overview

The author begins by evaluating ten candidate classifiers: Logistic Regression, Decision Tree, Random Forest, Extra Trees, KNN, AdaBoost, Gradient Boosting, XGBoost, LightGBM, and CatBoost, assessed on their accuracy, classification report, confusion matrix, and ROC-AUC score on training set. Based on this comparison, LightGBM and CatBoost are selected as the strongest performers and used in the final model. The training data is split into 10 folds, where in each fold both classifiers are trained on 9 folds and make predictions on the remaining fold. The predictions from LightGBM and CatBoost are then combined using $\hat{p} = \sqrt{0.70 \cdot p_{\text{LGB}}^2 + 0.30 \cdot p_{\text{CAT}}^2}$, giving LightGBM a higher weight. Final predictions on the test set are obtained by averaging these blended predictions across all 10 folds. Feature scaling is applied within each fold using `StandardScaler`, fitted only on the training portion to prevent data leakage.

3.3 Validation

The ADS is validated using the out-of-fold (OOF) ROC-AUC score. Because the model uses 10-fold cross-validation, each training instance is predicted exactly once by a model that did not train on it. Aggregating these predictions gives an unbiased estimate of how well the model generalizes to unseen data. The author reports a final OOF AUC of 0.926, indicating strong ability to distinguish between borrowers who repay and those who do not.

The author also submits predictions to the Kaggle leaderboard, which evaluates the model on a separate test set whose true labels are not publicly available. This model achieves an accuracy of 0.9326, providing an additional validation.

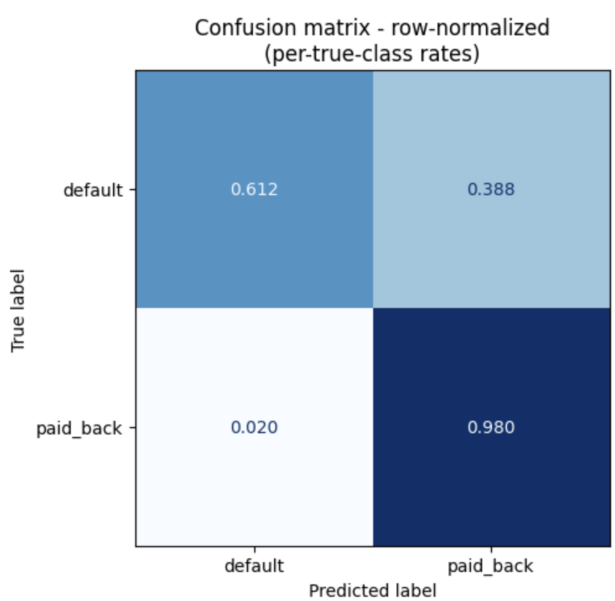
However, the validation is limited to predictive performance metrics, accuracy and AUC and does not assess fairness, calibration, or stability across demographic subgroups and intersectional subgroups. Meeting the stated goal of strong predictive performance does not imply that the model is appropriate for deployment in a real lending context, where regulatory and ethical requirements extend well beyond classification accuracy. Therefore, we are going to evaluate how this model behaves on the fairness metrics, and we will interpret how predictions are made in this model.

4 Outcomes

4.1 Accuracy Analysis for Baseline Model

The deployed ensemble achieves accuracy 0.906 and ROC-AUC 0.922, a meaningful improvement over the 0.799 majority-class base rate (10.7 pp gain). However, the per-class breakdown reveals a sharp asymmetry: recall on paid-back is 0.980 while recall on default is only 0.612, meaning the model correctly identifies 98% of true repayers but misses 39% of true defaulters (FPR = 0.388). The model approves 86.1% of all applicants at threshold 0.5.

This is a "loose approver" profile: generous to applicants and lenient on identifying default risk.



4.2 Fairness Analysis

To check whether the ADS treats demographic groups differently, we look at three protected attributes: `gender`, `marital_status`, and `education_level`, both on their own and in combination. We include the intersectional view because bias can hide in specific sub-groups even when the marginal numbers look fine.

4.2.1 Demographic Parity

Demographic parity asks that the positive prediction rate be roughly the same across groups. In this case, it's the probability of being predicted to repay. We compute the selection rate for each group at the default 0.5 threshold and compare.

The aggregate selection rate is 0.861, and the per-group rates cluster tightly around it. Disparate-impact ratios (min/max) are 0.997 across gender, 0.992 across marital status, and 0.970 across education level. Education shows the widest spread, with Bachelor's at 0.854 and PhD at 0.880. We read the higher PhD rate as calibration to the underlying repayment rate rather than preferential treatment, since the model is matching base rates that already differ by education. Overall, we don't see a demographic parity violation on the protected attributes in this dataset.

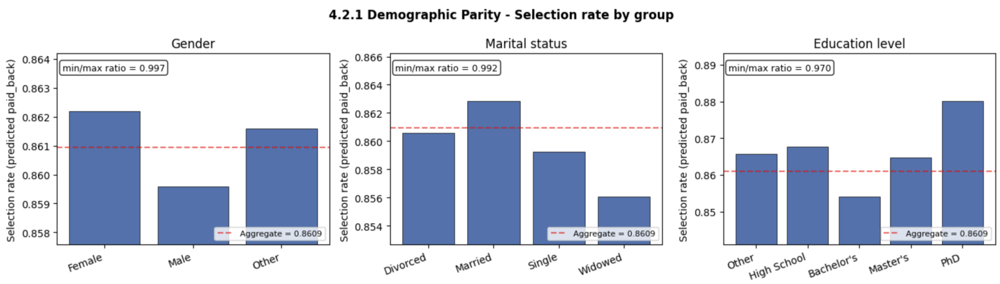


Figure 1: Selection rates by group for each protected attribute.

4.2.2 Equal Opportunity

Equal opportunity asks for uniform true positive rates (TPR) across groups: among borrowers who actually repaid, the probability of being correctly predicted to repay shouldn't depend on group membership.

TPR sits in a 0.35-percentage-point band across all groups (0.9777–0.9812). Disparate-impact ratios are 0.998 for gender, 0.998 for marital status, and 1.000 for education level, which is essentially perfect equality. Qualified applicants are identified at the same rate regardless of gender, marital status, or education level, so equal opportunity holds on every protected attribute we observe.

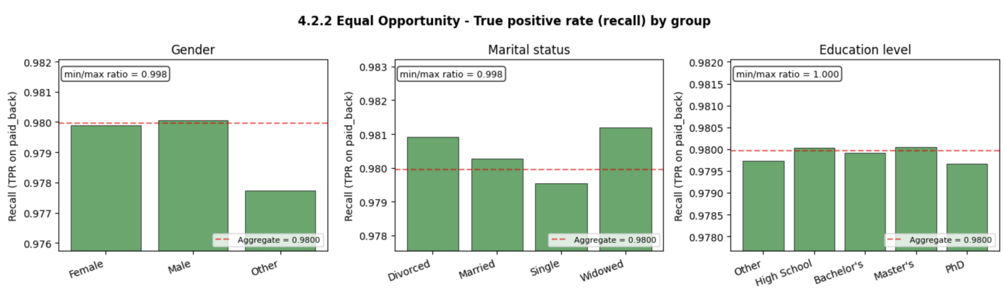


Figure 2: True positive rates by group for each protected attribute.

4.2.3 Error Rate Analysis

We also look at group-level false positive rates (FPR) and false negative rates (FNR). FPR is the probability the model predicts repayment for a borrower who doesn't actually repay; FNR is the probability it predicts non-repayment for a borrower who does.

FPR shows the largest disparities anywhere in our analysis. The disparate-impact ratios are 0.942 across gender (Female 0.386, Male 0.390, Other 0.410), 0.965 across marital status, and 0.955 across education level. The elevated FPR for the "Other" gender category (0.410, n = 3,728) and the "Other" education category (0.402, n = 26,677) is the main driver of cross-group variation. FNR, by contrast, is essentially uniform (0.019–0.022; ratios $\geq$ 0.895). So error rates are roughly equalized in both directions across the observed groups, and the only marginally concerning pattern is the slightly elevated FPR on the smaller minority categories.

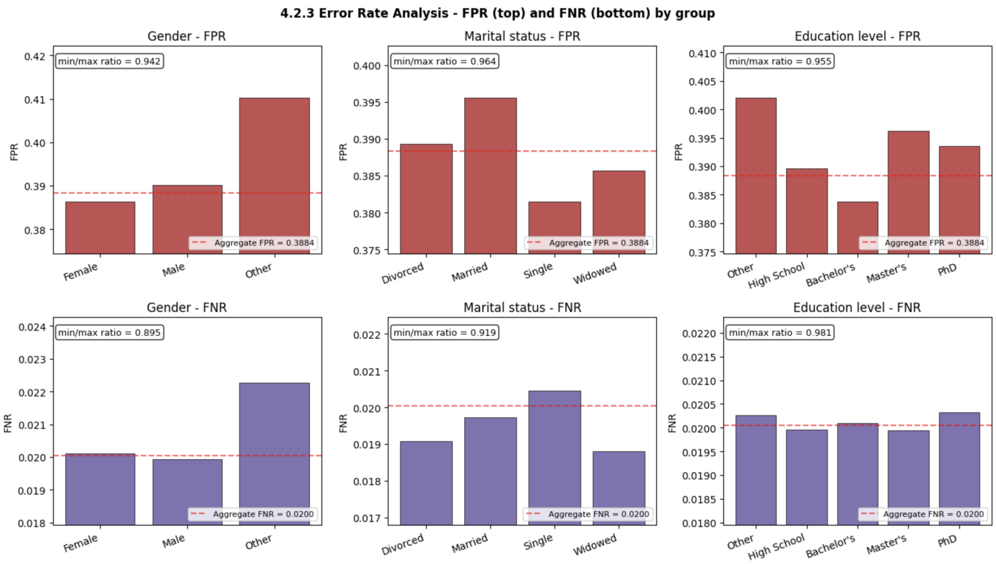


Figure 3: False positive and false negative rates by group for each protected attribute.

4.2.4 Intersectional Analysis

We also examine the pairwise intersections gender $\times$ marital status, gender $\times$ education level, and marital status $\times$ education level, restricted to cells with $n \geq 500$ to avoid noise from low-frequency combinations.

Intersectional disparities are larger than any single-attribute disparity but stay inside the 4/5ths threshold on selection rate (ratios 0.982, 0.943, 0.959 for the three intersections respectively) and on equal opportunity (0.992, 0.995, 0.992). Gender $\times$ education has the widest selection-rate spread: Female/PhD at 0.883 vs Other/Master's at 0.833, a 5.0 pp gap (ratio 0.943). Within marital $\times$ education, the extreme cells are Married/PhD (0.883) and Widowed/Bachelor's (0.847).

The intersectional pattern is consistent with calibration to underlying base rates rather than disparate treatment. The closest the analysis comes to flagging a concern is FPR on gender $\times$ education (ratio 0.847), driven mainly by the Other gender category and small-sample education tiers. This still clears the 4/5ths (0.80) threshold, but it's the widest intersectional gap of any metric in the analysis.

4.3 Additional Auditing Methods

4.3.1 Global Interpretability via SHAP

We compute global SHAP feature importance for both components of the loan-payback ensemble. The reported value per feature is the mean absolute SHAP value, averaged across the 10 fold-specific models on a 2,000-row sample. Features are ordered by LightGBM importance, and the same order is preserved for the CatBoost panel so the two are directly comparable. Two things stand out: employment-status-Unemployed dominates both components, and grade/subgrade do not appear among the top contributors.

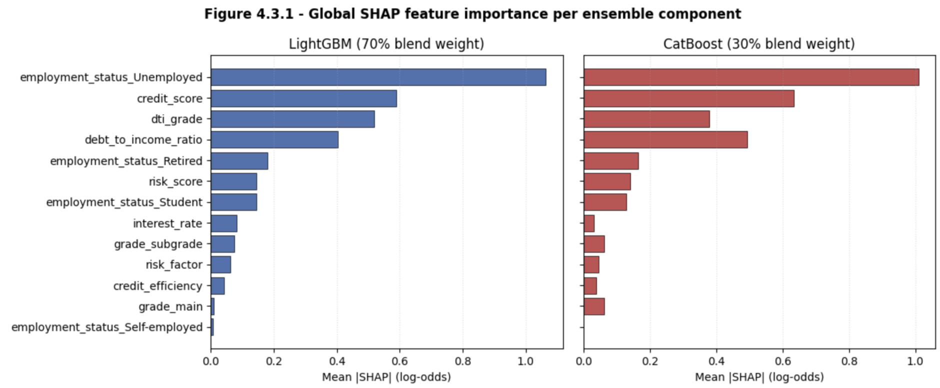


Figure 4: Global SHAP feature importance for the LightGBM and CatBoost components of the ensemble.

4.3.2 Local Interpretability via LIME

SHAP TreeExplainer is model-specific, and we applied it to LightGBM and CatBoost separately. To explain the full deployed ensemble, which combines the two components through a non-linear L2 blend over all 10 folds. We additionally use LIME, a model-agnostic local-explanation method, on the ensemble's `predict_proba`. We pick three representative borrowers from the training set by their out-of-fold predicted probability: a high-confidence repayer ($p = 1.000$), a high-confidence defaulter ($p = 0.000$), and a borderline case at $p = 0.500$ that actually defaulted. Figure 5 shows the top-10 LIME contributions for each.

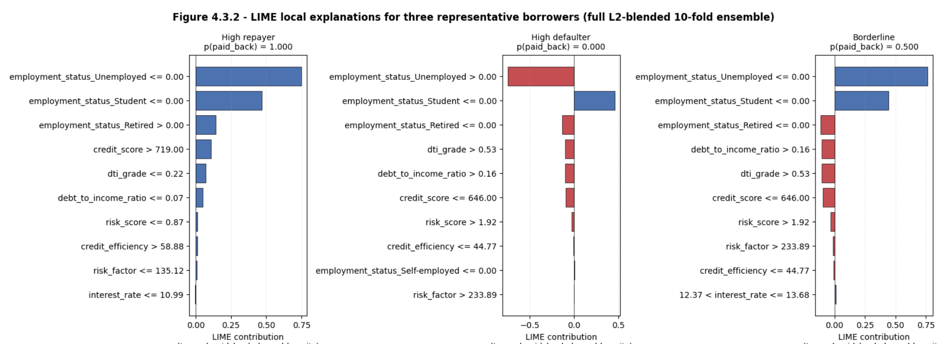


Figure 5: Top-10 LIME contributions for three representative borrowers.

The local explanations agree with the global SHAP picture. In all three cases the largest single contribution comes from employment-status-Unemployed: +0.74 (toward repayment) for the repayer, +0.76 (toward default) for the defaulter, and −0.75 (toward repayment) for the borderline case. The next-strongest features are employment-status-Student (+0.44 to +0.47) and employment-status-Retired (+0.12 to −0.13), again matching the global ranking. Conventional credit features such as debt-to-income ratio, DTI grade, credit score are present but contribute weights an order of magnitude smaller.

The borderline borrower (who actually defaulted) was classified just above the decision threshold almost entirely because they were not unemployed; once employment status is uninformative, the model has very little secondary signal to fall back on. SHAP and LIME come from different scopes (component-level global vs. ensemble-level local) and different mathematical foundations (Shapley values on tree paths vs. local linear regression on perturbations), and they arrive at the same conclusion: the ADS's predictions are dominated by employment-status proxies.

4.3.3 Stability and Robustness

We evaluate calibration with a reliability diagram, binning predictions and comparing each bin's mean predicted probability against the empirical repayment rate. The Brier score is 0.0716, a 55.4% reduction from the always-predict-base-rate baseline of 0.1607, indicating that the ensemble's probabilities are substantially more informative than chance and align well with observed outcomes.

We assess stability by perturbing 500 representative inputs with 5% Gaussian noise on the nine continuous features. Across 1,500 perturbed evaluations, the mean absolute shift in predicted probability is 0.029, and 95% of perturbations produce shifts below 0.122. The decision flip rate is 2.27%, so roughly 1 in 44 perturbed evaluations crosses the threshold. The flips are concentrated near the decision boundary, which is exactly where the LIME analysis showed the model relies almost entirely on employment-status indicators with weak secondary signals. So the flip rate is acceptable but worth noting alongside the interpretability finding.

5 Summary

5.1 Findings

Data appropriateness. The dataset is synthetically generated from a deep learning model trained on a real loan-prediction dataset, so the patterns it teaches are not the actual lending behavior. The data has no missing values, which is a a sign of synthetic origin rather than genuine data quality, and the target is imbalanced at roughly 80/20. Protected attributes (`gender`, `marital_status`, `education_level`) are present and used as features. For training a deployable system it is not acceptable: the synthetic origin leaves external validity unverified, and the inclusion of protected attributes raises legal exposure under disparate-impact standards regardless of the resulting metrics.

Robustness, accuracy, and fairness. The ensemble reaches accuracy 0.906 and ROC-AUC 0.922, but accuracy alone is misleading under class imbalance. The per-class breakdown shows a sharp asymmetry: recall on repayers is 0.980 while recall on defaulters is only 0.612, producing a "loose approver" profile that approves 86.1% of all applicants at threshold 0.5. Calibration is good (55.4% reduction from the base-rate baseline) and stability under 5% Gaussian noise is acceptable (mean shift 0.029, flip rate 2.27%). On fairness, demographic parity and equal opportunity are both satisfied at the single-attribute level (disparate-impact ratios ≥ 0.970 and ≥ 0.998 respectively). The substantive concern comes from the SHAP and LIME analysis: predictions are dominated by employment-status indicators, with conventional credit features contributing weights an order of magnitude smaller.

The chosen metrics map onto stakeholders as follows. **Lenders** care most about FPR (in the repay-as-positive frame, the rate at which true defaulters are predicted to repay), since approving a defaulter is direct financial loss; the model is weak here at 0.388. **Applicants** care most about FNR (being denied a loan they would have repaid), and the model is strong here at roughly 0.02. **Regulators and policymakers** care about the disparate-impact, which the model passes at the formal threshold. The disconnect between a formal fairness pass and the employment-proxy dominance found by SHAP/LIME is the finding we would flag to all three groups.

5.2 Deployment and Recommendations

We would not be comfortable deploying this ADS as-is, in either the public sector or industry. The decisive issues are not the formal fairness metrics, which mostly pass, but (1) the SHAP and LIME finding that predictions ride almost entirely on employment-status indicators, with traditional credit signals contributing an order of magnitude less; (2) the asymmetric error profile that approves 39% of true defaulters, which is the failure mode a credit model most needs to avoid; and (3) the synthetic origin of the training data, which leaves external validity unverified.

We recommend the following changes before any redeployment:

1. **Audit and constrain employment-status features.** Given that `employment_status_Unemployed` is the single largest contributor in both SHAP and LIME analyses, examine whether it is functioning as a legitimate credit signal or as a proxy that sidesteps the protected-attribute analysis.

2. **Tune the decision threshold against an asymmetric cost function.** The 0.5 default produces the loose-approver behavior; choosing the threshold on a cost-weighted utility would let the operator pick a point on the FPR/FNR frontier that matches their risk tolerance.

3. **Address class imbalance explicitly.** Use class-weighted training or resampling rather than the raw 80/20 split. This would also reduce the FPR/FNR asymmetry.
4. **Validate on real, non-synthetic data.** The synthetic generator has its own biases; any deployment claim requires hold-out evaluation on real lending records, ideally with demographic diversity beyond what the synthetic data provides.
5. **Document feature-selection decisions.** Several columns were dropped without stated rationale. These should be justified or revisited, particularly any operation that interact with the protected attributes.

6 Project Contributions

Simon's answer:

Simon: He was responsible for setting up and maintaining the LaTeX report structure throughout the project. He was also responsible for input and output profiling (Section 2) and the implementation and validation analysis (Section 3), including the investigation of encoding choices for `education_level`, the analysis of repayment rates and loan purpose distributions across education groups, and the description of the model pipeline and validation methodology. He also conducted the feature engineering examination and contributed to background and motivation (Section 1) and summary (Section 5).

Brian: He was responsible for the background and motivation (Section 1) and the outcomes section (Section 4). In Section 4, he conducted the baseline accuracy analysis including the confusion matrix and loose-approver profile characterization, the fairness analysis covering demographic parity, equal opportunity, error rate analysis, and intersectional analysis across gender, marital status, and education level, the global and local interpretability analysis using SHAP and LIME, and the stability and robustness evaluation including calibration and Gaussian noise perturbation tests. He also contributed to the findings and stakeholder mapping in the summary (Section 5).

Both of us jointly identified and selected the ADS from the Kaggle competition *Predicting Loan Payback*.

References

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[4] Nabiha Zahid. Loan prediction dataset 2025. `https://www.kaggle.com/datasets/nabihazahid/loan-prediction-dataset-2025/data`, 2025. Kaggle dataset. Accessed: 2026-03-11.